

Algorithmic Amplification

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1 Description of Artwork

Algorithmic Amplifications leverages the power of Large Language Models (LLMs) and a custom instant voice cloning pipeline to create a tangible representation of digital echo chambers, underscoring how language shapes our perceptions and reinforces our beliefs. It prompts a critical examination of the impact of AI on our linguistic and cultural landscapes, spotlighting the paradox of technological advancement: while LLMs mark a significant leap in AI, they also mirror and potentially amplify societal biases.

The rise of echo chambers on social media is frequently attributed to AI-driven recommender systems. These algorithms, designed to enhance user engagement, inadvertently create feedback loops. By prioritising content that aligns with a user's previous interactions, they limit exposure to diverse perspectives, reinforcing pre-existing beliefs. This mechanism of selective exposure often traps users inside ideological silos, unaware of the algorithmic influence shaping their information landscape. Social media companies have moved from their subscription- and network-based propagation algorithms towards increasingly algorithmic ones. It's not just what your friends say or share but also what the algorithm thinks you like best in order to drive engagement with the platform.

Algorithmic Amplifications is a conceptual artwork that physically manifests the abstract concept of social media echo chambers. Participants insert their upper body into a large enclosure suspended from the ceiling, symbolising the encapsulating nature of echo chambers. Upon entering, participants kick off the experience by expressing an opinion they hold. This opinion is then validated and reinforced by a chorus of synthetic voices that while discussing the issue at hand, merely echo and paraphrase the participant's opinion.

To technically achieve this we transcribe the audio input and instantly clone the voice of the participant. We then feed that input into a specially designed LLM pipeline that generates the dialogue of like-minded members of a mob. This interaction between agents



spawns an autonomous discussion between them that does not necessarily stay on point. Much like online discussions, as time goes by the arguments degenerate, occasionally go off-topic, become more extreme, more absolute, all without guidance. This dialogue is played back using a series of synthetic voices diverging gradually, in real-time, from the instantly cloned voice of the participant into increasingly twisted versions of the original voice. We enhance the feeling of a virtual world of opinions, by generating each voice from a specific point in space inside the sphere. This experience mimics the reinforcement encountered in digital echo chambers, where diverse opinions are drowned out by the overwhelming agreement and algorithmic amplification of one's own views. As soon as the user removes themselves from the echo chamber the voices fade out.

Algorithmic Amplification raises questions about what the future holds. Does the next step involve an even greater AI-driven reinforcement where synthetic minds crowd out human users, validating their views in order to drive public opinion and social media engagement. Inevitably, the questions that arise are about ownership and control of these algorithms that shape our narratives in the digital age.

2 Project URL

<https://www.randomquark.com/work/algorithmic-amplification>

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